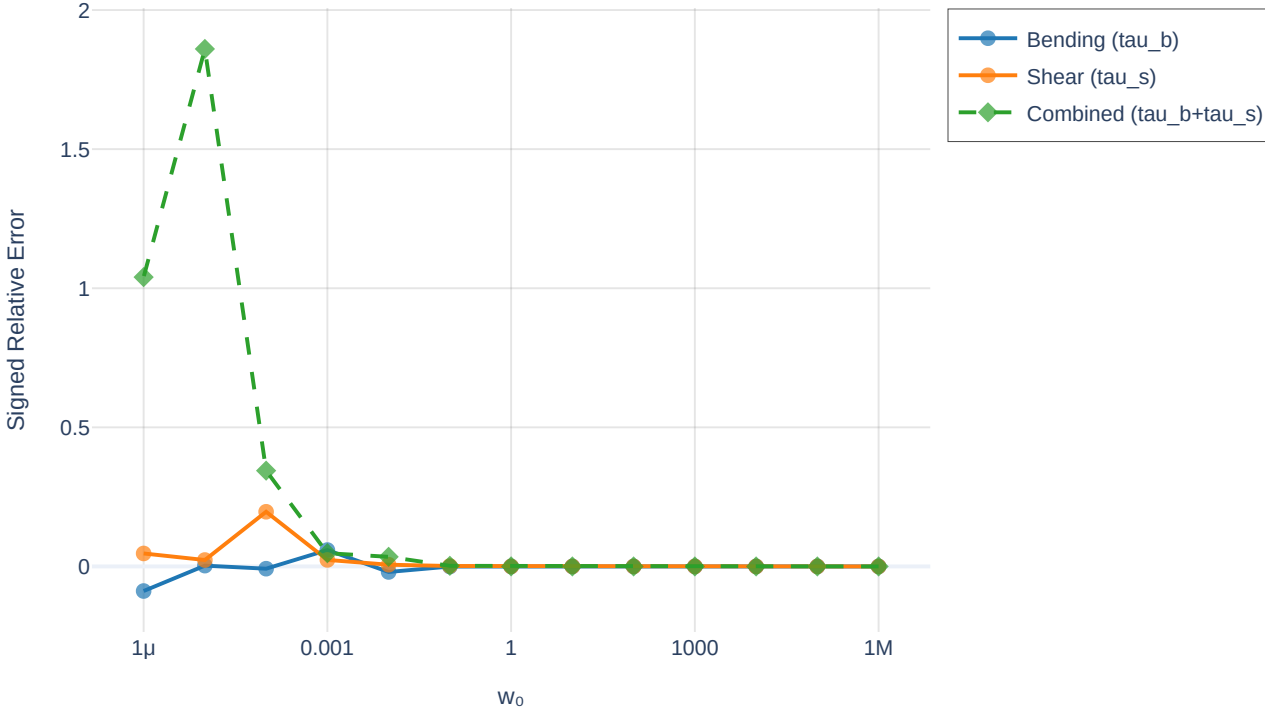Signed Relative Error vs External Parameter w_0